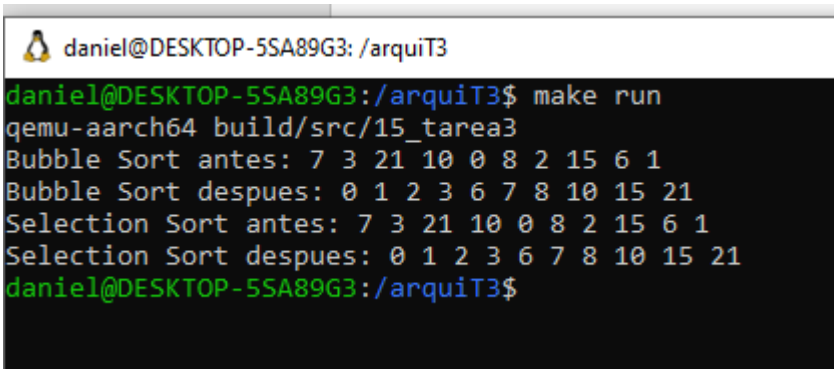

daniel@DESKTOP-5SA89G3: /arquiT3
daniel@DESKTOP-5SA89G3:/arquiT3$ make run
qemu-aarch64 build/src/15_tarea3
Bubble Sort antes: 7 3 21 10 0 8 2 15 6 1
Bubble Sort despues: 0 1 2 3 6 7 8 10 15 21
Selection Sort antes: 7 3 21 10 0 8 2 15 6 1
Selection Sort despues: 0 1 2 3 6 7 8 10 15 21
daniel@DESKTOP-5SA89G3:/arquiT3$

```
daniel@DESKTOP-5SA89G3:/arquiT3$ gdb-multiarch build/src/15_tarea3
GNU gdb (Ubuntu 17.1-2ubuntu1) 17.1
Copyright (C) 2025 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
Type "show copying" and "show warranty" for details.
This GDB was configured as "x86_64-linux-gnu".
Type "show configuration" for configuration details.
For bug reporting instructions, please see:
<https://www.gnu.org/software/gdb/bugs/>.
Find the GDB manual and other documentation resources online at:
    <http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word"...
Reading symbols from build/src/15_tarea3...
(gdb) set pagination off
(gdb) target remote :1234
Remote debugging using :1234

This GDB supports auto-downloading debuginfo from the following URLs:
  <https://debuginfod.ubuntu.com>
Enable debuginfod for this session? (y or [n]) n
Debuginfod has been disabled.
To make this setting permanent, add 'set debuginfod enabled off' to .gdbinit.
_start () at src/15_tarea3.s:145
145         ldr x1, =msg_bubble_before // direccion del mensaje
(gdb) break bubble_inner
Breakpoint 1 at 0x400144: file src/15_tarea3.s, line 65.
(gdb) break selection_inner
Breakpoint 2 at 0x4001a8: file src/15_tarea3.s, line 113.
(gdb) continue
Continuing.

Breakpoint 1, bubble_inner () at src/15_tarea3.s:65
65          cmp x4, x5              // comparar j con el limite interno
(gdb) info breakpoints
Num     Type           Disp Enb Address            What
1       breakpoint     keep y   0x0000000000400144 src/15_tarea3.s:65
        breakpoint already hit 1 time
2       breakpoint     keep y   0x00000000004001a8 src/15_tarea3.s:113
(gdb) info registers x0 x1 x2 x3 x4 x5 sp x29 x30
x0             0x410368            4260712
x1             0xa                 10
x2             0x0                 0
x3             0x9                 9
x4             0x0                 0
x5             0x9                 9
sp             0x7efdafdfdb10      0x7efdafdfdb10
x29            0x7efdafdfdb10      139628042509072
x30            0x400210            4194832
(gdb) x/10gd &array_bubble
0x410368:       7       3
0x410378:       21      10
0x410388:       0       8
0x410398:       2       15
0x4103a8:       6       1
(gdb)
```

```
(gdb) disable 1
(gdb) continue
Continuing.

Breakpoint 2, selection_inner () at src/15_tarea3.s:113
113         cmp x5, x1              // comparar j con la cantidad
(gdb) info breakpoints
Num     Type           Disp Enb Address            What
1       breakpoint     keep n   0x0000000000400144 src/15_tarea3.s:65
        breakpoint already hit 1 time
2       breakpoint     keep y   0x00000000004001a8 src/15_tarea3.s:113
        breakpoint already hit 1 time
(gdb) info registers x0 x1 x2 x3 x4 x5 sp x29 x30
x0             0x4103b8            4260792
x1             0xa                 10
x2             0x0                 0
x3             0x9                 9
x4             0x0                 0
x5             0x1                 1
sp             0x7adf1fffeb10      0x7adf1fffeb10
x29            0x7adf1fffeb10      135098733161232
x30            0x40024c            4194892
(gdb) x/10gd &array_selection
0x4103b8:       7       3
0x4103c8:       21      10
0x4103d8:       0       8
0x4103e8:       2       15
0x4103f8:       6       1
(gdb)
```